

Business Requirements Document Sieger Work Portal

Ticketing, Time Tracking, Approvals, Projects, Expense Claims, ABM CRM, Analytics and AI Integration

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This document captures the business and functional requirements of the Sieger Work Portal as implemented, records the business rules that govern it, and lists the open items that require a stakeholder decision. It is intended to serve as the agreed baseline for future change requests.

Document Control

Revision History

Version	Date	Author	Description of Change
0.1	1 Sep 2026	Business Analysis	Initial discovery: walkthrough of portal modules and workflows with Marketing and Service leads.
0.5	3 Sep 2026	Business Analysis	Functional requirements drafted per module; business rules catalogued from implemented behaviour.
1.0	4 Sep 2026	Business Analysis	Consolidated for review. Non-functional requirements, risks and open items added.

Approvals

Role	Name	Responsibility	Signature / Date
Business Owner	General Manager, Marketing	Approves scope, business rules and priorities	
Service Team Lead	Admin, Service	Confirms Service team requirements and categories	
Finance Representative	To be nominated	Confirms expense approval, verification and payment rules	
Technical Lead	Application Developer	Confirms feasibility, constraints and dependencies	

Distribution

Marketing leadership, Service leadership, Finance, IT and Application Development. This document is not for external circulation.

Reading Guide

Convention	Meaning
FR-xxx	Functional requirement. Numbered by module, traceable to a business rule or process step.
BR-xxx	Business rule. A policy the system must enforce regardless of which screen or channel triggers it.
NFR-xxx	Non-functional requirement covering security, performance, availability, auditability and usability.
OI-xxx	Open item requiring a stakeholder decision before the baseline can be closed.
Must / Should / Could	MoSCoW priority. "Must" requirements are live and relied upon daily. "Should" requirements are live but used by a subset of roles. "Could" requirements are live conveniences or partially delivered options.

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1. Executive Summary

The Sieger Work Portal is the internal operations system used by the Marketing and Service teams to raise, assign, track and close work. It replaced spreadsheet and email based tracking with a single portal that records who asked for work, who is doing it, how much time it was budgeted, how much time was spent, and whether the outcome was approved by a manager.

Over successive releases the portal has grown from a ticketing tool into a work management platform. It now covers:

- **Ticketing** with team-specific categories, priorities, divisions, time budgets and a full audit timeline.
- **Time tracking** against tickets, with consumed time rolled up for SLA reporting.
- **Approval workflows** for completion and for due date changes, routed to the requester's team admins.
- **Projects** that group tickets as tasks under a target date and division.
- **Recurring tasks** that generate the next occurrence automatically on a daily, weekly or monthly cadence.
- **Expense claims** with per-line receipt evidence, digitally signed approvals, printable verification codes and payment tracking.
- **Leave and permission** self-service records surfaced on the shared calendar.
- **Reporting:** analytics, completion and open-ticket reports, a downloadable PowerPoint work report and an expense spend report.
- **ABM CRM** for account-based outreach, with rule-driven pipeline progression and a daily action queue.
- **Advertising analytics** dashboards for Google Ads and LinkedIn.
- **Integrations:** Salesforce ticket intake, revocable API keys, a published OpenAPI specification and a Model Context Protocol (MCP) endpoint that lets AI assistants read and act on portal data under the connected person's own permissions.

A single organising principle runs through all modules: **every record belongs to a team, and every person sees only what their role on that team entitles them to see**. Super Admins span both teams. This document records that principle and every rule that flows from it, so that future changes can be assessed against an agreed baseline.

Purpose of this baseline. The portal was built incrementally without a consolidated requirements record. This BRD documents the requirements as they stand, flags the areas where business policy was never explicitly decided (Section 11), and provides the reference against which change requests will be raised and prioritised.

2. Business Context and Objectives

2.1 Problem Statement

Before the portal, work requests reached the Marketing team by email, chat and verbal instruction. There was no reliable record of what was asked, by whom, or when it was due. Time spent could not be compared against time allotted, so there was no objective basis for capacity planning or performance review. Expense reimbursement relied on paper bills and manual sign-off, which made verification by finance and auditors slow and error-prone. The Service team faced the same problems with a different vocabulary of work.

2.2 Business Objectives

Ref	Objective	Measure of Success
OBJ-1	Single system of record for all work requests across Marketing and Service	All requests, including those originating in Salesforce, exist as tickets with an owner, a due date and a status.
OBJ-2	Objective measurement of delivery against budget	Every completed ticket has allotted minutes, consumed minutes and a completion date; on-time rate and time variance are reportable per person, team and division.
OBJ-3	Managerial control over completion and deadline changes	No ticket reaches Completed or Closed by a non-admin without an admin decision; every deadline change on commissioned work is approved and recorded.
OBJ-4	Auditable, verifiable expense reimbursement	Every approved expense line carries evidence, an approver signature, a tamper-evident hash and a code that a third party can verify without portal access.
OBJ-5	Team isolation with shared platform	A Marketing admin never sees or changes Service records, and vice versa, while both use the same portal.
OBJ-6	Structured account-based outreach	Every ABM contact has a computed next action and due date; the daily queue is generated without manual planning.
OBJ-7	Data access for AI assistants and automation without widening permissions	External callers reach exactly the data the underlying person could reach, and no more.

2.3 Business Drivers

- Management visibility of workload, overdue work and pending approvals in real time.
- Reduction in reimbursement disputes through evidence-first, per-line approval.
- Compliance expectations from finance and auditors for signed, verifiable approvals.
- Growth of the Service team onto the same platform, requiring strict team separation.
- Adoption of AI assistants for reporting and routine actions, requiring a safe, permission-preserving access channel.

3. Scope

3.1 In Scope

Area	Included Capabilities
Access and administration	Login, password reset by one-time code, account enable/disable, role and designation management, team-scoped user administration, API key issuance.
Ticketing	Create, view, edit, assign, comment, change status, delete; team-specific categories; divisions; priorities; time budget; "Given By" attribution; audit timeline.
Time tracking	Log, edit and delete work sessions against a ticket; consumed time roll-up.
Approvals	Completion and waiting-state approval; due date change approval; pending approvals queue for admins.
Projects	Create and manage projects; group tickets as tasks; automatic timeline extension; division propagation; member notifications.
Recurring tasks	Daily, weekly and monthly recurrence with automatic generation and catch-up after downtime.
Views	Dashboard, ticket list, Kanban board, calendar (tickets, leave, permission), timeline/Gantt.
Notifications	In-app notification bell, email to admin on approval requests, real-time updates over WebSocket.
Expenses	Claims, line items, receipts, submission, per-line approval and rejection, resubmission, approve-all, payment marking, printable PDFs, public verification, spend report, signature management.
Leave and permission	Self-service recording of leave and short permission slots, visible on the calendar.
Reporting	Analytics dashboard, completion report, open tickets report, PowerPoint work report, expense report.
ABM CRM	Accounts, contacts, activities, opportunities, pipeline automation, daily queue, cadence settings, bulk import.
Advertising analytics	Google Ads campaign and keyword dashboards with AI insight panel; LinkedIn page analytics with OAuth connection.
Integrations	Salesforce inbound ticket webhook; OpenAPI specification; MCP endpoint with OAuth sign-in or API key; 40 AI-callable tools.

3.2 Out of Scope

- Payroll, HR master data and leave balance accounting. Leave and permission records are informational only.
- Payment execution. The portal records that an expense line was paid; it does not initiate bank transfers.
- Customer-facing ticketing or external customer logins.
- Salesforce outbound synchronisation (portal to Salesforce).
- Native mobile applications. The portal is a responsive web app installable as a PWA.

- Supabase Storage as a receipt store. The abstraction exists, but Google Drive is the only implemented backing store.

3.3 Solution Boundary

The solution comprises a web front end hosted on Vercel, a single API and WebSocket service hosted on Render, a Supabase Postgres database, Google Drive for binary files (receipts and signatures), and email delivery via SMTP with Resend as fallback. External parties are Salesforce (inbound), LinkedIn and Google Ads (analytics sources), Google Gemini and Anthropic (AI insight generation), and any MCP-capable AI client.

4. Stakeholders, Teams and User Roles

4.1 Stakeholders

Stakeholder	Interest	Involvement
General Manager, Marketing	Workload visibility, approval control, expense sign-off	Business owner; Super Admin; primary approver of Marketing expenses
Service Team Lead	Service ticket flow, categories, team isolation	Admin, Service; approver for Service team
Marketing and Service staff	Clear assignments, fair time budgets, quick reimbursement	Users and Team Members; ticket assignees and claimants
Finance	Verifiable approvals, approved-but-unpaid visibility	Consumer of expense report and verification codes; marks lines paid
Digital Marketing Executive	Advertising performance	Named user with Google Ads and LinkedIn access
ABM Executive	Outreach pipeline	Named user with ABM CRM access
Sales (Salesforce users)	Raising marketing requests from CRM	Source of webhook tickets
Auditors, banks, vendors	Authenticity of approval documents	Use the public verification page
Application Developer	Maintainability, security	Technical owner

4.2 Team Model

The portal serves two teams, **Marketing** and **Service**. Team membership is not a separate attribute; it is carried in the role label. This design keeps a single source of truth for both permission level and team. A role without a team suffix (legacy "Admin", "User", "Team Member") is treated as Marketing, because all users predating the Service rollout belonged to Marketing.

4.3 Role Definitions

Role	Team	Description and Typical Holder
Super Admin	All	Platform owner. Sees and manages both teams. The only role that can create or edit Super Admin accounts and assign across teams.
Admin - Marketing	Marketing	Marketing manager. Distributes work, sets time budgets, approves completion and deadlines, approves and pays Marketing expenses, manages Marketing users.
Admin - Service	Service	Service manager. Same authority as above, restricted to Service records and users.
User - MKTG	Marketing	Marketing executive. Raises tickets for own work, logs time, files expenses, may create projects.
User - Service	Service	Service executive. Same as above for Service.

Role	Team	Description and Typical Holder
Team Member - MKTG	Marketing	Assignable team member. Appears in the assignee list. Cannot create projects.
Team Member - Service	Service	Same as above for Service.

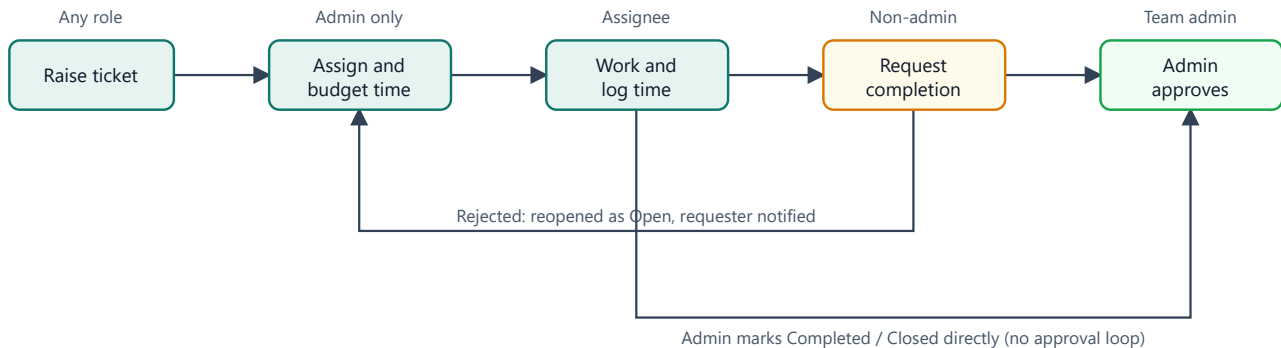
4.4 Access Matrix

Capability	Super Admin	Team Admin	User	Team Member
View tickets	All teams	Own team	Assigned or created by self	Assigned or created by self
Create ticket	Yes, assign anyone	Yes, assign within team	Yes, self-assigned	Yes, self-assigned
Assign / reassign ticket	Yes	Within team	No	No
Set allotted time, Given By	Yes	Yes	No	No
Mark Completed / Closed directly	Yes	Yes	Requests approval	Requests approval
Approve / reject completion	Yes	Own team	No	No
Change due date	Direct	Direct	Direct unless commissioned	Direct unless commissioned
Delete ticket	Yes	Own team	No	No
Log time	Any accessible ticket	Any team ticket	Own tickets	Own tickets
Create project	Yes	Yes	Yes	No
Edit project	Yes	Own team	Creator or owner	Creator or owner
File expense claim	Yes	Yes	Yes	Yes
Approve, reject, pay expense lines	Yes	Own team	No	No
Manage users	All	Own team, not Super Admins	No	No
Issue API keys	Yes	Yes	No	No
Google Ads, LinkedIn dashboards	Yes	Marketing admins only	Named user only	No
ABM CRM	Yes	Marketing admins only	Named users only	No
Completion and Open Tickets reports	Yes	Yes	No	No
Work Report (PPT)	Anyone	Own team	Self only	Self only

Capability	Super Admin	Team Admin	User	Team Member
Own signature, leave, permission	Yes	Yes	Yes	Yes

5. Business Process Overview

5.1 Ticket Lifecycle (Core Process)

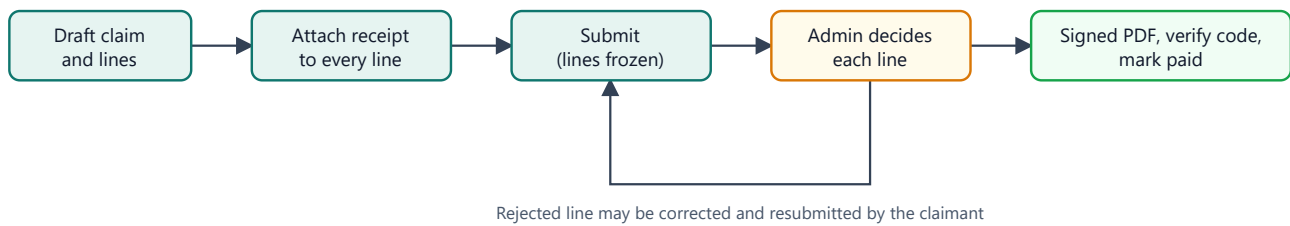


1. **Raise.** Any user raises a ticket with a title, description, priority, category, division and due date. A non-admin's ticket is assigned to themselves with no time budget. An admin may assign to any active person on their team and set the allotted minutes and the "Given By" attribution. Tickets may also arrive from Salesforce.
2. **Assign and budget.** An admin distributes work and sets the time budget that delivery will be measured against. Team admins may not hand work to the other team.
3. **Work and log time.** The assignee moves the ticket to In Progress and logs work sessions. Consumed minutes accumulate against the budget.
4. **Request completion.** When a non-admin sets Completed, Closed, Waiting For Sources or Waiting For Resources, the system checks time has been logged, places the ticket in Waiting For Approval and notifies the team's admins in-app and by email.
5. **Decide.** An admin approves (the requested status is applied and the completion date stamped) or rejects (the ticket returns to Open with the reason recorded). Admins changing status themselves bypass the loop.

5.2 Due Date Change Process

A non-admin asking for a new due date is handled in one of two ways. If the ticket was commissioned (has a "Given By") or the new date would push past the parent project's target date, the request is held as Pending, admins are notified and emailed, and the requester may withdraw it. Otherwise the change applies immediately with a timeline note. Admins change due dates directly. Any approved or direct date beyond a project's target automatically extends the project and informs its members.

5.3 Expense Claim Process



A claimant creates a claim, adds line items and attaches at least one receipt image or PDF per line. Submission freezes the lines and numbers them. A team admin with a registered signature approves or rejects each line individually. Approval produces a tamper-evident hash and a short verification code, and a signed PDF is available per line and per claim. Finance later marks approved lines as paid. The claim's status is always derived from its lines.

5.4 Recurring Task Process

When a ticket is created as recurring, it becomes the first occurrence and carries the "spawner" flag. An hourly scheduler generates the next occurrence when its date arrives, copies the template fields, hands the flag to the new occurrence and notifies the assignee. Missed periods during downtime are generated on catch-up, capped to prevent flooding.

5.5 ABM Outreach Process

An ABM user records target accounts and their contacts. Every touch (email, LinkedIn, WhatsApp, call, meeting, proposal) is logged as an activity. The activity type and result move the contact's pipeline status by fixed rules, and the next action and its due date are computed from a configurable cadence. The "Today's Queue" screen lists contacts whose next action is due, ranked by account tier and priority.

5.6 AI and Automation Access Process

An AI assistant such as ChatGPT or Claude connects to the MCP endpoint either by signing in as a person (OAuth) or with an API key acting as a person. Every tool call is executed through the portal's own API under that identity, so the assistant can read and change exactly what that person could, and every action is attributed to them in timelines and notifications.

6. Functional Requirements

Each requirement lists the actor, the requirement statement, the governing business rules and the acceptance criteria used to confirm it. Priorities follow MoSCoW as defined in the Reading Guide.

6.1 Authentication and Session Management

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-AUTH-01	Must	All users	Users authenticate with email and password. A successful login issues a session valid for 7 days.	Valid credentials return a session and the user's profile (name, role, division). Invalid credentials return a generic failure without revealing which field was wrong.
FR-AUTH-02	Must	All users	A disabled account cannot log in, and existing sessions and API keys for that account stop working.	Login for a disabled account is refused with "Account disabled"; API keys acting as that user are refused within 30 seconds.
FR-AUTH-03	Must	All users	Users can reset a forgotten password by requesting a 6-digit one-time code to their registered email, valid for 10 minutes with a maximum of 5 attempts.	Code is delivered by email; expired or exhausted codes are rejected; a successful reset replaces the password and invalidates the code.
FR-AUTH-04	Should	Admin	An admin can trigger a password reset email on behalf of a user in their team.	The user receives the one-time code; the admin never sees the code.
FR-AUTH-05	Must	System	Login and reset endpoints are rate limited per IP and per target email to prevent credential stuffing and inbox flooding.	Login: 20 attempts per 15 minutes per IP. Reset request: 10 per 15 minutes per IP and 5 per hour per email.
FR-AUTH-06	Must	System	When a session expires or is rejected, the web app clears local state, disconnects real-time updates and returns the user to the login page.	A single expired session triggers exactly one redirect even when several requests fail simultaneously.
FR-AUTH-07	Could	All users	The web app is installable as a Progressive Web App on desktop and mobile.	Manifest and icons are served; the app can be added to the home screen.

6.2 User and Role Administration

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-USR-01	Must	Admin	Admins can create users with name, email, password, role, division and designation.	Email and password are mandatory; the created user can log in immediately.
FR-USR-02	Must	Admin	Admins can enable or disable a user, change their role, division and designation, and reset their password.	Changes take effect on the next request; disabling stops all access as per FR-AUTH-02.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-USR-03	Must	System	A team admin may only view and manage users on their own team. Super Admin accounts are visible to all admins for assignment purposes but manageable only by a Super Admin.	A Marketing admin attempting to edit a Service user or create a Super Admin is refused with a clear message.
FR-USR-04	Must	System	Designation (job title) is set by an admin, never by the user, because it is printed above the user's signature on approved expense documents.	No self-service path exists to edit designation.
FR-USR-05	Must	System	Password hashes are never returned to any client.	All user listings return only the agreed profile columns.
FR-USR-06	Must	Admin	Admins can view the list of assignable team members scoped to their team when creating tickets.	Only active Team Member roles on the caller's team are offered; Super Admins see both teams.

6.3 Ticket Management

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-TKT-01	Must	All users	Any user can create a ticket with title, description (both mandatory), priority, category, division, due date and optional project.	Ticket is created in Open status with a "Ticket created" timeline entry and the creator recorded.
FR-TKT-02	Must	System	A ticket raised by a non-admin is assigned to that user regardless of any assignee submitted, and starts with zero allotted minutes.	Non-admin create requests with a different assignee still result in self-assignment.
FR-TKT-03	Must	Admin	An admin can assign a ticket at creation or later to any active user on their team, and set allotted minutes and "Given By".	Assignee's display name is stored on the ticket; assignee receives a notification; disabled users cannot be chosen.
FR-TKT-04	Must	System	Ticket categories are team-specific. Marketing and Service each have their own list, and a ticket only accepts categories from its team's list.	An invalid category for the ticket's team is rejected with a message naming the team. Legacy tickets keep their historical category.
FR-TKT-05	Must	All users	Users can view the tickets they are entitled to see, with assignee, creator, team, status, due date and logged time entries.	Non-admins see tickets assigned to or created by them; team admins see their team; Super Admins see all.
FR-TKT-06	Must	All users	Users with access can update title, description, priority, division, category and status, and add comments.	Every change appends a timeline entry naming the actor and time.
FR-TKT-07	Must	System	Only an admin can change allotted minutes or "Given By".	A non-admin attempting either change is refused.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-TKT-08	Must	All users	Ticket statuses available to users are Open, In Progress, Waiting For Sources, Completed and Closed. Waiting For Approval is system-set only.	The status picker never offers Waiting For Approval.
FR-TKT-09	Must	System	A ticket cannot move to Completed, Waiting For Sources or Waiting For Resources unless at least one time entry has been logged.	Attempt without time entries is refused with "Please log time before marking ticket as completed".
FR-TKT-10	Must	System	A ticket marked Completed receives a completion date; reopening clears it.	Completion date equals the IST date of approval or direct completion; null after reopen.
FR-TKT-11	Must	Admin	An admin can reassign a ticket within their team. Reassignment that would move the ticket to the other team is refused.	Previous and new assignees both receive real-time updates; new assignee is notified.
FR-TKT-12	Must	Admin	An admin can delete a ticket on their team.	Non-admins are refused; the ticket and its time entries are removed.
FR-TKT-13	Must	All users	Every ticket carries a complete, chronological timeline of creation, assignment, status changes, comments, time logs, approvals, due date events and project links.	Timeline entries are immutable and visible on the ticket detail page.
FR-TKT-14	Should	All users	Users can close a ticket with a closure reason appended to the description.	Closure reason is visible after closure; non-admin closure follows the approval loop.
FR-TKT-15	Should	Admin	Admins can edit a ticket's core fields on a dedicated edit page.	Edit page is reachable only by admins.
FR-TKT-16	Must	System	The ticket list shows whether the assignee is an active account, so reports can exclude work owned by disabled people.	Unassigned tickets remain visible; tickets held by disabled users are flagged.

6.4 Time Tracking

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-TIME-01	Must	Assignee, Admin	Users can log a work session against a ticket they can access, with work date, start time, end time, duration in minutes and notes.	Entry is stored with the logger's name; ticket consumed minutes increase by the duration; timeline records "logged Xh Ym".
FR-TIME-02	Must	System	Duration must be a non-negative number.	Invalid or negative durations are rejected.
FR-TIME-03	Must	All users	Users can view all time entries for a ticket they can access, newest first.	Entries list logger, date, times, duration and notes.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-TIME-04	Must	Owner, Admin	The person who logged an entry, or an admin over the ticket's team, can edit or delete it. Consumed minutes are adjusted accordingly.	Others are refused; consumed minutes never go below zero.
FR-TIME-05	Must	System	Consumed minutes against allotted minutes are visible on the ticket and used in analytics and SLA reporting.	Variance and utilisation are derivable from the two figures.

6.5 Completion and Due Date Approval Workflows

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-APR-01	Must	System	When a non-admin sets a ticket to Completed, Closed, Waiting For Sources or Waiting For Resources, the ticket enters Waiting For Approval with the requested status recorded, and the team's admins are notified in-app and by email.	Status shows Waiting For Approval; requested status and requester are stored; admins of the requester's team (and Super Admins) receive a notification; the configured admin mailbox receives an "Approval Required" email with a link to the ticket.
FR-APR-02	Must	Admin	An admin can approve a pending completion request. The requested status is applied, the approver and time are recorded, and the requester is notified.	Ticket status equals the requested status; completion date set when Completed; timeline shows "Ticket approved".
FR-APR-03	Must	Admin	An admin can reject a pending completion request. The ticket returns to Open and the requester is notified.	Timeline shows "Ticket rejected"; rejected-by and time are recorded.
FR-APR-04	Must	System	An admin setting any of these statuses applies it directly without an approval loop, and the assignee is notified if different from the admin.	No Waiting For Approval state is entered for admin actions.
FR-APR-05	Must	Admin	Admins have a Pending Requests screen listing all completion and due date requests awaiting decision on their team, with approve and reject actions.	Both request types appear with requester, current and requested values and one-click actions.
FR-APR-06	Must	Non-admin	A non-admin can request a new due date. The request requires approval only if the ticket has a "Given By" or the new date exceeds the parent project's target date; otherwise it applies immediately.	Commissioned or project-exceeding requests are stored as Pending with requester and time; others change the date at once with a timeline note explaining why no approval was needed.
FR-APR-07	Must	Admin	An admin can approve or reject a pending due date request. Approval applies the requested date; both outcomes notify the requester.	Only one pending request per ticket; approval clears the request fields; timeline records the outcome.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-APR-08	Should	Non-admin	The requester can withdraw their own pending due date request.	Request fields are cleared; timeline shows "cancelled by requester".
FR-APR-09	Must	System	Due date approval requests are emailed to the configured admin mailbox with current date, requested date, requester and a review link.	Email is sent asynchronously and never delays the user's response.
FR-APR-10	Must	System	Approving or directly setting a due date beyond the parent project's target date automatically extends the project's target date and notifies project members.	Project target date equals the new due date; members receive "Project Timeline Extended".

6.6 Projects and Task Grouping

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-PRJ-01	Must	Admin, User	Admins and Users can create a project with name (mandatory), description, target date, colour, division, owner and members. Team Members cannot create projects.	Project is Active on creation; members are notified "Added to Project".
FR-PRJ-02	Must	All users	Users see projects they own, created, belong to, or have a task in. Team admins see their team's projects; Super Admins see all.	Project list is filtered accordingly and shows per-project task statistics.
FR-PRJ-03	Must	All users	The project detail page shows the project, its member details and all tasks (tickets) with status, assignee and due date, and supports drag-and-drop status changes and in-context task creation.	Task changes made on the project page follow all ticket rules and approval loops.
FR-PRJ-04	Must	Admin, Owner, Creator	The project can be edited by an admin on its team, its owner or its creator.	Others are refused.
FR-PRJ-05	Must	System	A task's due date may not exceed the project target date unless set by an admin, in which case the project timeline extends automatically.	Non-admin attempt is refused naming the target date; admin attempt extends the project.
FR-PRJ-06	Must	System	A project's target date cannot be set earlier than the latest due date of any of its tasks.	Attempt is refused naming the latest task due date.
FR-PRJ-07	Must	System	A project's division propagates to all its tasks; new tasks inherit the project's division.	Changing project division updates every linked ticket.
FR-PRJ-08	Should	Admin, Owner	Existing tickets can be linked to or unlinked from a project.	Timeline records "Added to project" or "Removed from project".

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-PRJ-09	Should	Admin	Deleting a project unlinks its tasks; it does not delete tickets.	Tickets remain with project reference cleared.
FR-PRJ-10	Should	System	Ownership transfer notifies the new owner.	"Project Ownership" notification received.
FR-PRJ-11	Must	System	Project progress statistics distinguish "late" (all tasks done after target) from "overdue" (open tasks past target).	Completion percentage and days late are computed and shown.

6.7 Recurring Tasks

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-REC-01	Must	All users	A ticket can be created as recurring with an interval of daily, weekly or monthly.	The first occurrence's title is suffixed with its period (e.g. "Week of 4 Aug 2026") and its due date is one interval ahead.
FR-REC-02	Must	System	The system generates the next occurrence automatically when its scheduled date arrives, copying description, priority, category, division, assignee, budget, "Given By" and project from the template.	Occurrence is created in Open with a "Recurring task auto-created" timeline entry; assignee is notified.
FR-REC-03	Must	System	Generation runs shortly after service start and hourly thereafter, and catches up on periods missed during downtime, capped at 60 occurrences per series per run.	After an outage, missed occurrences appear within an hour of restart.
FR-REC-04	Must	System	Exactly one occurrence exists per series and period, even if two service instances run concurrently.	Duplicate generation is prevented by claim-before-insert and a database uniqueness rule.
FR-REC-05	Should	All users	Every occurrence displays a "recurring" badge and its series base title.	Badge visible in lists and detail.
FR-REC-06	Should	System	Monthly recurrence preserves the day of month and clamps to month end for shorter months, recovering to the original day when possible.	31 Jan, then 28 Feb, then 31 Mar.

6.8 Work Views: Dashboard, Kanban, Calendar, Timeline

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-VIEW-01	Must	All users	A dashboard summarises the user's visible tickets: counts by status and priority, overdue work, upcoming due dates and recent activity, with team-appropriate branding.	Figures reconcile with the ticket list; Service users see Service branding.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-VIEW-02	Must	All users	A Kanban board shows tickets in status columns with drag-and-drop status changes.	Dragging to a column updates status subject to all ticket rules; board refreshes in real time.
FR-VIEW-03	Must	All users	A calendar shows tickets by due date together with leave and permission records.	Month, week and day views; entries link to the ticket.
FR-VIEW-04	Should	All users	A timeline (Gantt) view shows tickets across time and provides the entry point for recording leave and permission.	Bars reflect created and due dates; leave and permission forms available.
FR-VIEW-05	Must	All users	The ticket list supports filtering and searching by status, priority, category, division, assignee and text.	Filters combine; results reflect the user's visibility scope.
FR-VIEW-06	Should	All users	Navigation groups dashboards, reports, ABM and admin functions, and hides items the user cannot access.	No dead-end links for any role.

6.9 Notifications and Real-Time Updates

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-NTF-01	Must	All users	Users receive in-app notifications for assignment, approval requests and decisions, due date events, project membership and timeline changes, recurring task creation and expense decisions.	Notification bell shows unread count; each item links to the related ticket where applicable.
FR-NTF-02	Must	All users	Users can mark a notification as read.	Unread count decreases; state persists.
FR-NTF-03	Must	System	Notifications are addressed to a user identity, not a display name, so renaming a user or two users sharing a name does not misroute notifications.	Renamed users retain their notification history.
FR-NTF-04	Must	System	Approval requests to admins are limited to the requester's team's admins plus Super Admins.	A Service request never notifies a Marketing admin.
FR-NTF-05	Must	System	Ticket, project and notification changes are pushed in real time to connected users who are entitled to see them, using the same visibility rules as the API.	Open pages update without refresh; a Marketing user never receives a Service event.
FR-NTF-06	Must	System	Email is sent for completion and due date approval requests and for password reset codes. Email delivery must never delay or fail the originating action.	Failures are logged; the user action succeeds regardless.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-NTF-07	Must	System	User-supplied text in emails (ticket titles, names) is rendered as text, never as active content.	A title containing HTML renders literally.

6.10 Expense Claims

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-EXP-01	Must	All users	Any user can create an expense claim with a title, division and currency (default INR). The claim's team is frozen from the claimant's role at creation.	Claim is created in Draft; team cannot be changed later.
FR-EXP-02	Must	Claimant	The claimant adds line items with expense date, category, description, amount and tax amount.	Date, category and positive amount are mandatory; tax cannot be negative; category must belong to the claim's team list.
FR-EXP-03	Must	System	Claim total is always recomputed from lines (amount plus tax) and never accepted from the client.	Total equals the sum of line gross amounts after every change.
FR-EXP-04	Must	Claimant	Each line can carry one to five receipts (images or PDF, up to 5 MB each, up to 25 per claim). Receipts are stored outside the database and streamed back only to authorised viewers.	File type is validated by content, not extension; receipt is downloadable from the claim page.
FR-EXP-05	Should	System	A receipt identical to one already on another claim is flagged as a possible duplicate but not blocked.	Warning shown naming the other claim.
FR-EXP-06	Must	Claimant	A claim can be submitted only when it has at least one line and every line has a receipt. Submission freezes and numbers the lines and notifies the team's admins.	Missing receipts are listed by line; lines are numbered 01, 02...; status becomes Submitted.
FR-EXP-07	Must	Claimant	Lines can be added to a submitted claim; they go directly to approvers as Pending. Approved lines cannot be edited or deleted.	Editing an approved line is refused.
FR-EXP-08	Must	System	Only the claimant can change their claim. Admins on the claim's team and Super Admins can view it. Teammates may view and contribute lines where allowed.	Others are refused.
FR-EXP-09	Must	All users	Users can list their claims with status and totals; admins can list their team's claims.	List filterable by status and claimant.
FR-EXP-10	Should	Claimant	A claim can be deleted by its claimant while no line has been approved.	Deletion removes lines and receipts.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-EXP-11	Must	System	Claim status is derived from its lines: Draft, Submitted, Partially Approved, Approved, Rejected or Paid.	Status is recomputed after every line decision or payment.

6.11 Expense Approval, Signature, Verification and Payment

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-EXA-01	Must	All users	Each user can upload, view and remove their own approval signature image.	No user can set another user's signature.
FR-EXA-02	Must	Admin	An admin on the claim's team (or a Super Admin) can approve an individual submitted line, provided the admin has a registered signature and the line has at least one receipt.	Missing signature returns "Upload your signature before approving"; missing receipt returns "cannot be approved".
FR-EXA-03	Must	System	On approval the system records approver identity, name, role, designation and time, computes a SHA-256 hash over the claim, line, receipts and approver, and derives a 10-character verification code excluding ambiguous characters.	Hash and code are stored on the line; code is unique system-wide.
FR-EXA-04	Must	System	A claim number in the form EXP-YYYY-NNNN is minted on the first decision on the claim, never for abandoned drafts.	Numbers are sequential within a year.
FR-EXA-05	Must	Admin	An admin can reject a line with a reason. The claimant is notified.	Reason is stored and displayed.
FR-EXA-06	Must	Claimant	A rejected line can be corrected and resubmitted for approval.	Line returns to Pending; previous rejection reason retained in timeline.
FR-EXA-07	Should	Admin	An admin can approve all pending lines on a claim in one action. Each line still receives its own hash and code; lines without receipts are skipped and reported.	Response lists approved and skipped lines with reasons.
FR-EXA-08	Must	Admin	An admin can mark an approved line as paid, reverse that marking, or mark all approved unpaid lines paid.	Pending or rejected lines cannot be paid; payer and time recorded; approval record untouched.
FR-EXA-09	Must	All users	A signed PDF is available per approved line and per claim, showing the line, its receipts, the approver's signature, designation, date and the verification code.	PDF renders the frozen designation and role as at approval.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-EXA-10	Must	Public	Anyone holding a printed verification code can confirm the document is genuine on a public page without logging in. The response contains only the reference, claimant, team, amount, category, date, approver and time.	Unknown, unapproved or altered codes return "not valid" identically; no other data is exposed; rate limited to 30 checks per minute.
FR-EXA-11	Must	System	Legacy claim-level verification codes printed before per-line approval continue to resolve.	Old printouts verify.
FR-EXA-12	Must	System	Every approval, rejection, resubmission and payment appends to the claim's timeline and notifies the claimant.	Timeline is complete and ordered.

6.12 Leave and Permission Requests

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-LV-01	Should	All users	Users can record leave with type, from date, to date and reason.	Record is stored against the user and appears on the calendar.
FR-LV-02	Should	All users	Users can record a short permission slot with date, from time, to time and reason.	Record appears on the calendar.
FR-LV-03	Should	Admin	Admins see all leave and permission records; other users see only their own.	Filtering applied by role.
FR-LV-04	Could	System	Records are stored with status Approved on creation. No approval workflow exists. See OI-04.	Status is informational only.

6.13 Reporting

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-RPT-01	Must	All users	An analytics page shows ticket metrics (volume, status mix, priority mix, on-time rate, allotted versus consumed time, per-person and per-division breakdowns), scoped to the viewer's visibility.	Non-admins see their own tickets only.
FR-RPT-02	Must	Admin	A completion report lists completed tickets over a period with on-time status, budget variance and assignee, filterable by division and person.	Exportable; excludes disabled users' work by default.
FR-RPT-03	Must	Admin	An open tickets report lists open and overdue work by assignee, priority and age.	Overdue defined as due date before today (IST) and not Completed or Closed.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-RPT-04	Should	All users	A Work Report is downloadable as a PowerPoint deck for self, a chosen person or the whole team, over a date window by created or due date, with a preview of contents before download.	Non-admins can only run their own; API keys cannot export a whole-team report; rate limited to 20 per 5 minutes; empty windows return "nothing to report".
FR-RPT-05	Must	All users	An expense report totals spend by claimant, division, category and month, with buckets for approved, paid, unpaid, pending and rejected, each showing net, tax and gross.	Paid and unpaid are subsets of approved; figures reconcile to line items; scope follows viewer's rights.
FR-RPT-06	Should	All users	A reports hub page links to all reports the user may access.	No inaccessible links shown.

6.14 Account-Based Marketing (ABM) CRM

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-ABM-01	Must	System	ABM is available to Marketing admins and specifically named Marketing users. Service team users never have access.	Others are refused on both navigation and API.
FR-ABM-02	Must	ABM user	Users manage accounts with name, country, industry, website, size, revenue, priority, tier, status, owner, division and notes.	Account statuses: Research, Contacting, Engaged, Meeting Scheduled, Proposal, Negotiation, Won, Lost, Nurturing.
FR-ABM-03	Must	ABM user	Users manage contacts under an account with designation, role level, email, phone, LinkedIn, WhatsApp, decision-maker flag, pipeline status, do-not-contact flag and manual next action override.	Pipeline status is held on the contact, not the account.
FR-ABM-04	Should	ABM user	Contacts can be imported in bulk and their status updated in bulk.	Import reports created and skipped rows.
FR-ABM-05	Must	ABM user	Every touch is logged as an activity with type, channel, result, notes and date against an account and optionally a contact.	Activity types and results as listed in Section 8.
FR-ABM-06	Must	System	Logging an activity updates the contact's pipeline status by rule (e.g. "Cold Email 1" sets "Email 1 Sent"; result "Meeting Fixed" sets "Meeting Scheduled"; "Not Interested" sets "Future Follow-up").	Status transitions match the rule table in BR-ABM-01.
FR-ABM-07	Must	System	The next action and due date for each contact are computed from the last activity and configurable wait days, unless a manual override exists or the contact is closed, wrong or do-not-contact.	Sequence: Email 1, Email 2, LinkedIn Connect, LinkedIn Message, WhatsApp, Call, Follow-up; positive responses jump to "Schedule Meeting".

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-ABM-08	Must	ABM user	A "Today's Queue" lists contacts whose next action is due, ranked by account tier and priority weights.	Tier 1 and High priority rank first.
FR-ABM-09	Should	Admin	Sequence wait days per activity type are editable in settings within 0 to 365 days.	Defaults apply when unset.
FR-ABM-10	Should	ABM user	Opportunities are recorded per account with name, potential, probability, expected value and stage.	Stages: Qualification, Proposal, Negotiation, Closed Won, Closed Lost.
FR-ABM-11	Must	ABM user	An ABM dashboard summarises accounts by status, tier and division, activity volume and positive outcomes.	Figures reconcile with underlying records.

6.15 Advertising Analytics Dashboards

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-ADS-01	Should	Marketing admin, named user	A Google Ads dashboard presents spend, clicks, impressions, conversions, CPC, CTR, conversion rate and cost per conversion, with campaign and keyword analyses, trend charts, efficiency matrices, waste analysis and an executive scorecard.	Access limited per the access matrix; data sourced from imported campaign analysis tables.
FR-ADS-02	Could	Marketing admin	An AI insight panel generates narrative summary and recommendations from the dashboard data.	Insight is labelled as AI-generated.
FR-ADS-03	Should	Marketing admin, named user	A LinkedIn dashboard connects to the company page via OAuth and presents follower, post and engagement analytics.	Token stored server-side; expired connection prompts reconnection.
FR-ADS-04	Must	System	Service team users never have access to advertising dashboards.	Navigation hidden and API refused.

6.16 Integrations: Salesforce, API Keys, OpenAPI and MCP

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-INT-01	Must	Salesforce	Salesforce can create a Marketing ticket by posting title, description, priority, division and due date to a webhook secured by a shared secret.	Missing or wrong secret is refused; ticket is created in Open, attributed to "Salesforce", and broadcast to Marketing admins in real time; 60 requests per minute limit.
FR-INT-02	Must	Admin	Admins can issue API keys that act as a chosen user, with a name, optional expiry and a read-only flag. The key is shown once.	Key acts with exactly that user's permissions; read-only keys are refused any write; revocation takes effect within 30 seconds.

ID	Priority	Actor	Requirement	Acceptance Criteria
FR-INT-03	Must	System	API keys cannot be issued or revoked by a machine caller; only a signed-in person may do so.	Key and MCP sessions are refused on key management endpoints.
FR-INT-04	Should	Integrators	A public OpenAPI specification describes the tickets, projects, expenses and users endpoints.	Specification is retrievable without credentials and contains no data.
FR-INT-05	Must	AI client	An MCP endpoint allows an AI assistant to connect by OAuth sign-in (access for one hour, renewable for thirty days) or by API key.	Sign-in page is hosted by the portal; the assistant never sees the password; disabling the account ends access within 30 seconds.
FR-INT-06	Must	AI client	The MCP endpoint exposes 14 read tools (whoami, tickets, statistics, projects, expense report and claims, users, table discovery and query, search and fetch) and 26 write tools (tickets, time, approvals, projects, expense claims and lines, users).	Every tool executes through the portal API as the connected person; write tools are refused unless write access was granted at connection.
FR-INT-07	Must	System	The MCP layer never bypasses portal permissions and never exposes passwords, API key hashes or LinkedIn tokens.	A team member's connection sees only their own work; the three excluded tables cannot be named.
FR-INT-08	Should	AI client	Both modern streamable HTTP and legacy SSE transports are supported; results are trimmed and summarised server-side to stay within AI context limits.	Oversized results return guidance to narrow the query rather than truncated data.

7. Business Rules Catalogue

Business rules apply on every channel: web app, API key, Salesforce webhook and MCP. They are enforced once, in the API, and never restated in a client.

7.1 Team and Access Rules

ID	Rule	Rationale
BR-TEAM-01	Every role carries its team. Super Admin spans all teams. Roles without a team suffix are Marketing.	Single source of truth for permission and team.
BR-TEAM-02	A ticket belongs to the team of its assignee; if unassigned, the team of its creator; if neither, Marketing.	Deterministic ownership for scoping and real-time audience.
BR-TEAM-03	A team admin sees and acts on their own team's tickets, projects, claims and users only.	Team isolation (OBJ-5).
BR-TEAM-04	A non-admin sees tickets assigned to them or created by them.	Creators keep access to what they raised.
BR-TEAM-05	An assignment that would move a ticket to another team is refused for team admins. Assignment to a Super Admin is permitted only if the ticket stays in the admin's team.	Prevents silent re-homing across teams.
BR-TEAM-06	A project belongs to the team of its creator, falling back to its owner, then Marketing.	Consistent with ticket rule.
BR-TEAM-07	An expense claim's team is frozen from the claimant's role at creation and never changes.	A later role change must not move approved spend between teams' books.
BR-TEAM-08	Service team users have no access to Google Ads, LinkedIn or ABM regardless of role.	Business decision at Service rollout.

7.2 Ticket and Approval Rules

ID	Rule	Rationale
BR-TKT-01	Only an admin distributes work. A non-admin's ticket is always self-assigned.	Managers control workload.
BR-TKT-02	Only an admin sets allotted minutes and "Given By".	Assignees must not grade their own budget or remove the approval trigger.
BR-TKT-03	Default priority is Medium. Category is optional. Division is free text on tickets.	Low friction on creation.
BR-TKT-04	Category must belong to the ticket's team list when set or changed; historical values are tolerated on unrelated edits.	Team-specific vocabularies without blocking old tickets.
BR-TKT-05	Completed, Closed, Waiting For Sources and Waiting For Resources require admin approval when set by a non-admin.	OBJ-3.

ID	Rule	Rationale
BR-TKT-06	Completed, Waiting For Sources and Waiting For Resources require at least one logged time entry.	No completion without evidence of work.
BR-TKT-07	A due date change by a non-admin requires approval only if the ticket has a "Given By" or the date exceeds the project target date.	Commissioned work and project timelines are protected; own work is flexible.
BR-TKT-08	Only one due date request may be pending per ticket; the requester may withdraw it.	Unambiguous decision queue.
BR-TKT-09	Rejection of a completion request returns the ticket to Open.	Work continues.
BR-TKT-10	Completion date is set when Completed is applied and cleared on reopen.	Accurate completion reporting.
BR-TKT-11	Only admins delete tickets.	Audit protection.
BR-TKT-12	All timestamps are recorded in Indian Standard Time. "Today" for overdue and completion is the IST date.	Single business timezone.

7.3 Project Rules

ID	Rule	Rationale
BR-PRJ-01	Team Members cannot create projects.	Projects are planning constructs owned by executives and managers.
BR-PRJ-02	Task due dates may not exceed the project target date unless set by an admin, which extends the project.	The project timeline follows its tasks under managerial control.
BR-PRJ-03	The project target date cannot be earlier than the latest task due date.	Consistency between plan and tasks.
BR-PRJ-04	Project division applies to all its tasks.	Consistent reporting by division.
BR-PRJ-05	Deleting a project unlinks tasks; it never deletes tickets.	Work records are preserved.

7.4 Expense Rules

ID	Rule	Rationale
BR-EXP-01	Every line must carry a receipt before submission and before approval.	Evidence-first reimbursement (OBJ-4).
BR-EXP-02	Approval is per line. The claim's status is a roll-up of its lines and is never decided directly.	Envelope can never disagree with contents.
BR-EXP-03	An approver must be an admin on the claim's team (or Super Admin) and must have a registered signature.	Every approval bears a signature.
BR-EXP-04	An admin may approve a claim on which they are the claimant. The record (approver, role, designation, hash) makes this visible. See OI-06.	Deliberate business decision; segregation replaced by attribution.

ID	Rule	Rationale
BR-EXP-05	The approval hash covers the receipt file digests. Replacing a receipt after approval invalidates the printed code.	Tamper evidence.
BR-EXP-06	Approver designation and role are frozen onto the line at approval.	A later promotion must not rewrite an old document.
BR-EXP-07	Approved lines cannot be edited or deleted.	Protects the signed record.
BR-EXP-08	Only approved lines can be marked paid. Payment is additive and does not alter the approval record.	"Approved but unpaid" is the figure finance chases.
BR-EXP-09	A claim is Paid only when every approved line is paid and every line has been decided; rejected lines do not hold it back.	Sensible settlement semantics.
BR-EXP-10	Claim numbers (EXP-YYYY-NNNN) are minted on first decision, never for drafts.	No gaps from abandoned drafts.
BR-EXP-11	Divisions on claims must be one of ASTOR, CPS, TMD, All User, or blank.	Spend reporting integrity.
BR-EXP-12	Public verification returns identical "not valid" for unknown, unapproved and altered codes and exposes no identifiers.	Codes are short; guesses must be worthless.

7.5 ABM Pipeline Rules

ID	Rule	Rationale
BR-ABM-01	Status after activity: Research sets Contact Identified; Cold Email 1/2/3 set Email 1/2/3 Sent; LinkedIn Connect sets LinkedIn Requested (or Connected on result Connected); LinkedIn or Sales Navigator Message sets LinkedIn Message Sent; WhatsApp sets WhatsApp Sent; Direct Call sets Called; Meeting sets Meeting Scheduled; Proposal Sent sets Proposal Sent. Results override: Meeting Fixed sets Meeting Scheduled; Responded or Interested sets Interested; Not Interested sets Future Follow-up; Wrong Contact sets Wrong Contact. Follow-up, Demo and Visit leave status unchanged.	Consistent pipeline without manual status editing.
BR-ABM-02	Default wait days: emails 5, LinkedIn Connect 5, LinkedIn Message 3, WhatsApp 3, Call 5, Follow-up 5, Meeting and Demo 1, Proposal 7, Visit 3. Editable in settings.	Cadence discipline.
BR-ABM-03	No next action is computed for closed, wrong or do-not-contact contacts, or after Not Interested. A manual next action always wins.	Respect outcomes and human judgement.
BR-ABM-04	Queue ranking weights: Tier 1 = 15, Tier 2 = 5, Tier 3 = 0; High = 10, Medium = 4, Low = 0.	Focus on the most valuable accounts.

7.6 Integration and Automation Rules

ID	Rule	Rationale
BR-INT-01	An API key is a standing grant of one user's permissions and can never exceed them. It is refused if the user is disabled, the key expired or was revoked.	Attributable, revocable automation.
BR-INT-02	Read-only keys and read-only AI sessions are refused every write.	Narrowest useful scope.

ID	Rule	Rationale
BR-INT-03	Machine callers cannot mint or revoke keys.	Revoking a connection must end its access.
BR-INT-04	MCP tools call the portal's own API; no permission rule is re-implemented in the AI layer.	A second copy of a rule is the one that drifts.
BR-INT-05	Salesforce tickets are Marketing tickets with no assignee, budget zero, created by "Salesforce".	Enter the normal triage flow.

8. Data Requirements and Reference Data

8.1 Key Business Entities

Entity	Key Attributes	Relationships
User	Name, email, role, division, designation, active flag, signature reference	Creates and is assigned tickets; owns claims; approves
Ticket	Title, description, priority, category, division, status, due date, completed date, allotted minutes, consumed minutes, Given By, approval fields, due date request fields, recurrence fields, timeline	Belongs to a user (assignee), optionally a project; has time entries
Time Entry	Work date, start, end, duration minutes, notes, logged by	Belongs to a ticket
Project	Name, description, status, target date, colour, division, owner, members, creator	Has many tickets as tasks
Notification	Recipient, title, message, related ticket, read flag	Addressed to a user
Expense Claim	Claim number, claimant, team, title, division, currency, total, status, revision, timeline	Has many lines
Expense Line	Line number, date, category, description, amount, tax, approval status, approver, approver role and designation, approved at, hash, verification code, rejection reason, paid at, paid by	Belongs to a claim; has receipts
Receipt	Storage reference, file name, type, size, content hash, uploaded by	Belongs to a line and claim
Leave / Permission Request	User, type or date, from, to, reason, status	Belongs to a user
API Key	Name, prefix, hash, acting user, creator, read-only, expiry, revoked, last used	Acts as a user
ABM Account	Name, country, industry, website, employees, revenue, priority, tier, status, owner, division, notes	Has contacts, activities, opportunities
ABM Contact	Name, designation, role level, channels, decision maker, status, do-not-contact, next action override, last activity summary	Belongs to an account
ABM Activity	Type, channel, result, notes, date, created by	Belongs to an account and contact
ABM Opportunity	Name, potential, probability, expected value, stage	Belongs to an account

8.2 Reference Data

List	Values
Ticket statuses	Open, In Progress, Waiting For Sources, Waiting For Resources, Waiting For Approval (system), Completed, Closed

List	Values
Ticket priorities	Low, Medium (default), High, Critical (normalised in reports from historical free-text values)
Marketing ticket categories	Animation, Branding, Campaign, Collateral, Content, Email Campaign, Exhibition, Others, Reports, Salesforce, Social Media, Strategy, Video, Website
Service ticket categories	Spares / Logistics Follow up, Customer Follow up, Sales Force, Reports / MIS, Site / Installation, Breakdown Support, Rotary, AMC Follow up, Training, Contractor / Vendor - Coordination, Project Updates, Sales - Coordination
Divisions (tickets, projects, expenses)	ASTOR, CPS, TMD, All User
Recurrence intervals	Daily, Weekly, Monthly
Marketing expense categories	Ad - Digital, Ad - Magazine, Customer Engagement, Content Creation, Exhibition, Print Collaterals, Software Subscriptions, Merchandise, Travel - Air / Rail, Local Conveyance, Accommodation, Meals, Photography / Videography, Agency / Freelancer, Training / Conference, Courier / Postage, Telephone & Internet, Contingency
Service expense categories	Travel - Air / Rail, Local Conveyance, Accommodation, Meals, Fuel, Spares Purchase, Tools & Consumables, Courier / Freight, Site Expenses, Telephone & Internet, Training, Other
Expense claim statuses	Draft, Submitted, Partially Approved, Approved, Rejected, Paid
Expense line statuses	Pending, Approved, Rejected; display status Paid when paid
Currency	INR (default)
ABM divisions	ASTOR - TEX, ASTOR - NON TEX, ILS, CPS, TMD EXP
ABM account statuses	Research, Contacting, Engaged, Meeting Scheduled, Proposal, Negotiation, Won, Lost, Nurturing
ABM contact statuses	Research Pending, Contact Identified, Email 1 Sent, Email 2 Sent, Email 3 Sent, LinkedIn Requested, LinkedIn Connected, LinkedIn Message Sent, WhatsApp Sent, Called, Interested, Meeting Scheduled, Proposal Sent, Negotiation, Closed Won, Closed Lost, Future Follow-up, Wrong Contact
ABM activity types	Research, Cold Email 1, Cold Email 2, Cold Email 3, Follow-up, LinkedIn Connect, LinkedIn Message, Sales Navigator Message, WhatsApp, Direct Call, Meeting, Proposal Sent, Demo, Visit
ABM activity results	Sent, Opened, Clicked, Connected, Responded, No Response, Interested, Not Interested, Wrong Contact, Meeting Fixed
ABM tiers, priorities, role levels	Tier 1, Tier 2, Tier 3; High, Medium, Low; Decision Maker, Influencer, Gatekeeper, Individual Contributor
ABM opportunity stages	Qualification, Proposal, Negotiation, Closed Won, Closed Lost

8.3 Data Retention and Ownership

- Tickets, time entries, claims and lines are retained indefinitely as the audit record. Deletion is admin-only and is not soft-delete. See OI-08.
- Receipts and signature images are held in a private Google Drive folder owned by the organisation's Gmail account; the database stores only the file reference and content hash. The folder must never be

link-shared.

- Passwords are stored as bcrypt hashes; API keys as SHA-256 hashes; one-time codes are held in memory for 10 minutes only.

9. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All API and real-time access requires a valid session or API key. The service refuses to start if the signing secret or database credentials are absent.
NFR-02	Security	Browser access is limited to the approved portal origins. Key-based and MCP callers are exempt because they carry no ambient credentials.
NFR-03	Security	Authorization is enforced in the API for every record read or written. No client-side check is relied upon.
NFR-04	Security	Secrets compared on inbound webhooks use constant-time comparison. Random codes and keys use cryptographically secure generation.
NFR-05	Security	Error responses never expose stack traces or internal paths. Malformed or oversized request bodies return a 4xx with a plain message.
NFR-06	Security	Uploaded files are validated by content signature; PDFs are probed for structure. Executables are rejected.
NFR-07	Rate limiting	Login, password reset, verification, Salesforce webhook, API key creation, work report generation and MCP calls are rate limited as specified in their requirements.
NFR-08	Auditability	Every material change to a ticket or claim is recorded in an immutable timeline with actor and IST timestamp. Approval records include approver identity, role, designation and a cryptographic hash.
NFR-09	Data integrity	Listings of more than 1,000 rows are paged completely; no report may silently truncate. Claim totals are recomputed from lines on every write.
NFR-10	Availability	The API runs as a single instance on a hosted platform that may sleep when idle. First request after idle may take up to a minute. Unhandled errors are logged without terminating the service.
NFR-11	Performance	Common list views return within 3 seconds for a team of 20 with up to 10,000 tickets. MCP reads are cached for 20 seconds per caller and path to avoid repeated heavy queries.
NFR-12	Real time	Ticket, project and notification updates reach entitled connected users within 2 seconds of the change.
NFR-13	Resilience to schema drift	The service continues to operate when an optional database migration has not yet been applied, degrading the affected feature with a clear message that names the migration to run.
NFR-14	Email	Email is delivered via SMTP with automatic fallback to a transactional provider; delivery failures never block user actions.
NFR-15	Usability	The portal is responsive on desktop and mobile, installable as a PWA, and presents team-specific branding and vocabulary.
NFR-16	Localisation	All business dates and times are Indian Standard Time; currency defaults to INR.
NFR-17	Interoperability	The API is described by an OpenAPI document; the MCP endpoint supports protocol versions 2024-11-05, 2025-03-26 and 2025-06-18.

ID	Category	Requirement
NFR-18	Privacy	Public endpoints (verification, OpenAPI, OAuth discovery) expose no personal data beyond what is printed on the verified document.

10. Assumptions, Constraints and Dependencies

10.1 Assumptions

- Both teams operate in Indian Standard Time and reimburse in INR.
- Each person has one role and therefore one team; nobody works across teams except Super Admins.
- Managers (admins) are the appropriate approvers for both work completion and expenses on their team.
- Email addresses in the user directory are current and monitored, since password reset depends on them.
- Salesforce is the only external system creating tickets, and it creates Marketing tickets only.
- The organisation's Google account used for Drive storage remains active with sufficient quota.

10.2 Constraints

- The database has no row-level security; all access control is in application code. Any new data access path must go through the API.
- Schema changes are applied by running SQL scripts manually in the database console. There is no automated migration pipeline.
- The API runs as a single instance. In-memory state (one-time codes, key lookup cache, OAuth authorisation codes, MCP response cache) is not shared across instances.
- Google Drive is accessed as a personal Gmail account via a refresh token; the OAuth consent screen must remain published to production to prevent token expiry every 7 days.
- The database REST interface caps responses at 1,000 rows and does not support transactions; read-modify-write sequences (e.g. consumed minutes) are not atomic.

10.3 Dependencies

Dependency	Purpose	Impact if unavailable
Supabase (Postgres)	All business data	Portal unavailable
Render	API and WebSocket hosting	Portal unavailable; cold starts after idle
Vercel	Web front end hosting	Web app unavailable; API and MCP still up
Google Drive	Receipts and signatures	Cannot upload or view receipts; approvals blocked
SMTP / Resend	Approval and reset emails	In-app notifications continue; password reset blocked
Salesforce	Inbound tickets	Sales requests must be raised manually
LinkedIn API, Google Ads data	Analytics dashboards	Dashboards stale or unavailable
Gemini / Anthropic	AI insight generation	Insight panel unavailable; dashboards unaffected

11. Risks and Open Items

The following items were identified during analysis. Each requires a business or technical decision before it can be closed. Recommended owners are indicated.

11.1 Open Items for Stakeholder Decision

ID	Item	Recommendation	Owner
OI-01	Access to ABM, Google Ads and LinkedIn is granted to specific named individuals by identifier rather than by a role or permission flag. Staff changes require a code change.	Introduce a per-user feature permission set managed in the Admin Panel.	Business Owner
OI-02	Legacy roles without a team suffix still exist and are treated as Marketing by convention.	Run the role normalisation script and retire legacy labels.	Technical Lead
OI-03	Approval and reset emails go to a single configured admin mailbox rather than to the specific team's admins.	Route emails to the admins actually notified in-app, per team.	Business Owner
OI-04	Leave and permission requests are auto-approved and have no admin decision, no balance and no conflict check.	Decide whether these remain informational or become an approval workflow with HR ownership.	Business Owner, HR
OI-05	Ticket division is free text, while expense and project divisions are validated lists.	Validate ticket division against the same list and backfill historical values.	Business Owner
OI-06	An admin may approve a claim on which they are the claimant. Segregation was intentionally removed; the record provides attribution instead. Expense lines do not record who added them.	Confirm with Finance. If segregation is required, record line authorship and reinstate the rule at that level.	Finance
OI-07	"Waiting For Resources" triggers approval but is not offered in the status picker; "Waiting For Sources" is.	Confirm the intended waiting states and align the picker and rules.	Business Owner
OI-08	Ticket and project deletion is permanent with no recycle bin.	Consider soft-delete with admin restore for tickets carrying logged time.	Business Owner
OI-09	Two recurring series sharing the same base title and period date would collide.	Enforce unique base titles per assignee, or widen the uniqueness rule.	Technical Lead
OI-10	Only Google Drive is implemented as a receipt store; Supabase Storage is a stub.	Either complete the alternative store or remove the option to avoid misconfiguration.	Technical Lead
OI-11	The MCP write surface allows AI clients to create users and delete records under an admin's identity.	Agree which write tools should require an explicit human confirmation step or be excluded by policy.	Business Owner, IT

11.2 Risks

ID	Risk	Likelihood	Impact	Mitigation
RSK-01	Single API instance; scaling out would break in-memory codes, caches and OAuth state.	Low	High	Move shared state to the database or a cache service before scaling.
RSK-02	Concurrent time logging on one ticket may lose an update to consumed minutes.	Low	Medium	Recompute consumed minutes from entries on read, or move the increment to a database function.
RSK-03	Drive refresh token expiry if the consent screen is moved back to testing.	Low	High	Document the setting; monitor upload failures.
RSK-04	Cold start delay on the hosting platform frustrates the first user of the day and can time out receipt uploads.	High	Low	Keep-alive ping or paid tier.
RSK-05	Manual SQL migrations may be skipped in an environment, leaving features degraded.	Medium	Medium	Maintain a migration checklist; the service already reports which script is missing.
RSK-06	Notifications for users sharing a display name fall back to name matching.	Low	Low	Run the notification backfill and enforce unique display names.
RSK-07	A key placed in a connector URL (path-based fallback) may leak through logs.	Medium	Medium	Use short-expiry keys for that mode and revoke on disconnect, as documented.

12. Glossary

Term	Definition
ABM	Account-Based Marketing. Targeted outreach to named accounts and their contacts.
Allotted minutes	The time budget an admin assigns to a ticket; the basis for delivery measurement.
Consumed minutes	The sum of logged time entries on a ticket.
Claim	An expense claim: an envelope of line items filed by one claimant.
Division	A business unit used for reporting: ASTOR, CPS, TMD, All User.
Given By	The person who commissioned a ticket. Its presence makes due date changes subject to approval.
IST	Indian Standard Time, the portal's single business timezone.
Line	One expense on a claim, approved and paid individually.
MCP	Model Context Protocol. A standard by which AI assistants call tools exposed by an application.
Occurrence	One generated instance of a recurring task.
PWA	Progressive Web App; a web app installable on a device home screen.
Spawner	The most recent occurrence of a recurring series, responsible for generating the next.
Team admin	Admin - Marketing or Admin - Service.
Timeline	The immutable, ordered history of events on a ticket or claim.
Verification code	A 10-character code printed on an approved expense document, derived from its approval hash, checkable publicly.
Waiting For Approval	System-set ticket status while a non-admin's completion request awaits an admin decision.

Appendix A: State Models

A.1 Ticket Status Transitions

From	To	Actor	Condition and Effect
(new)	Open	Any, Salesforce, Scheduler	Creation.
Open	In Progress	Assignee, Admin	Direct; timeline entry.
Open, In Progress	Waiting For Approval	Non-admin	Requested Completed, Closed, Waiting For Sources or Waiting For Resources; time logged (except Closed); admins notified.
Waiting For Approval	Requested status	Admin	Approve; completion date set if Completed; requester notified.
Waiting For Approval	Open	Admin	Reject; requester notified.
Any	Completed, Closed, Waiting states	Admin	Direct; recorded as approved by the admin; assignee notified.
Completed	Open, In Progress	Admin, Assignee	Reopen; completion date cleared.

A.2 Due Date Request States

State	Meaning
(none)	No request outstanding.
Pending	Non-admin requested a date on commissioned or project-exceeding work; admins notified.
Approved	Admin applied the requested date; project extended if needed.
Rejected	Admin declined; original date stands.
Reverted	Requester withdrew their own pending request.

A.3 Expense Claim and Line States

Claim Status	Derived When
Draft	Not yet submitted; lines editable; no receipts required yet.
Submitted	Submitted and no line decided yet (or lines added later still pending).
Partially Approved	Some lines decided with a mix of outcomes, or some still pending after at least one decision.
Approved	Every line approved.
Rejected	Every line rejected.
Paid	Every line decided, at least one approved, and every approved line paid.

Line Status	Transitions
Pending	On creation. Approve to Approved; reject (with reason) to Rejected.
Approved	Frozen. May be marked Paid and reversed. Never editable.
Rejected	Claimant may edit and resubmit to Pending.
Paid (display)	Approved with payment recorded.

A.4 ABM Contact Pipeline

Research Pending, Contact Identified, Email 1 Sent, Email 2 Sent, Email 3 Sent, LinkedIn Requested, LinkedIn Connected, LinkedIn Message Sent, WhatsApp Sent, Called, Interested, Meeting Scheduled, Proposal Sent, Negotiation, Closed Won, Closed Lost. Side states: Future Follow-up (after Not Interested) and Wrong Contact. Transitions are driven by logged activities per BR-ABM-01.

Appendix B: Interface Summary

The following summarises the functional interfaces of the portal for traceability. Paths are relative to the API base.

Area	Interface	Purpose
Authentication	POST /auth/login; /auth/forgot-password; /auth/reset-password; /auth/admin-send-reset; GET /auth/team-members	Session issue, password recovery, assignee list
Users	GET, POST /users; PUT, DELETE /users/:id; /users/me/signature	Administration and personal signature
Tickets	GET, POST /tickets; GET, PUT, DELETE /tickets/:id; PUT /tickets/:id/approve, /reject, /assign	Lifecycle, approvals, assignment
Time entries	POST /ticket-time-entries; GET /ticket-time-entries/:ticketId; PUT, DELETE /ticket-time-entries/:id	Work logging
Projects	GET, POST /projects; GET, PUT, DELETE /projects/:id; GET /projects/meta/members	Project management
Notifications	GET /notifications; PUT /notifications/:id/read	In-app notifications
Expenses	GET, POST /expenses; /expenses/:id and lines, receipts, submit, approve, reject, resubmit, approve-all, pay, unpay, pay-all, pdf; GET /expenses/report; GET /expenses/meta/categories	Claims end to end
Verification	GET /verify/:code (public)	Document authenticity
Leave and permission	GET, POST /leave-requests; GET, POST /permission-requests	Self-service records
Reports	GET /reports/subjects; /reports/work.pptx; /reports/work/preview	PowerPoint work report
ABM	/abm/dashboard, /queue, /settings, /accounts, /contacts, /contacts/bulk, /activities, /opportunities	CRM
Advertising	/google-ads/*; /linkedin/*	Analytics dashboards
Salesforce	POST /salesforce/mkt-ticket (shared secret)	Inbound tickets
API keys	GET, POST /api-keys; DELETE /api-keys/:id	Machine credentials

Area	Interface	Purpose
OpenAPI	GET /openapi.json (public)	API description
MCP	POST /mcp; /mcp/sse; /mcp/k/:key; OAuth discovery, register, authorize, token	AI assistant access
Real time	WebSocket events: ticketCreated, ticketUpdated, projectCreated, projectUpdated, notificationReceived	Live updates to entitled users

Traceability note. Functional requirements in Section 6 map to these interfaces one-to-one. Any change request should cite the FR and BR identifiers it affects, so that the impact on other channels (API key, Salesforce, MCP) is assessed alongside the web app.

End of document.